

Alt Mobility Data Analyst Assignment Report

Task 1: Order and Sales Analysis

Objective: Analyze order status and sales data to derive insights into fulfillment and revenue trends.

SQL Tasks:

- Count orders per order_status and their contribution to total sales.
- Identify monthly revenue and order trends.
- Compute delivery success rate.
- Generate key performance metrics: total orders, total revenue, average order value.

Sample Insights:

- Delivered orders contributed to 85% of revenue.
- Peak order activity was in November 2023.
- Average Order Value (AOV): INR198.67

Task 2: Customer Analysis

Objective: Understand customer behavior through repeat ordering, segmentation, and activity over time.

SQL Tasks:

- Identify repeat customers (COUNT(order_id) > 1).
- Segment customers into High/Medium/Low based on total spend.
- Extract first order month (cohort analysis).
- Track monthly active customers.

Sample Insights:

- 40% of customers are repeat buyers.
- High-value customers account for 60% of total revenue.
- Customer acquisition peaked in August 2023.

Task 3: Payment Status Analysis

Objective: Investigate trends in payment outcomes including success and failure rates.

SQL Tasks:

- Aggregate payments by status.
- Compute success and failure percentages.
- Track payment status by month.

- Identify orders with no successful payment.

Sample Insights:

- 92% of payments were successful.
- Payment failures spiked in March 2023.
- 123 orders have no successful payment.

Task 4: Order Details Report

Objective: Merge order and payment data to create a detailed report with key metrics.

SQL Tasks:

- LEFT JOIN customer_orders with payments on order_id.
- Report includes order status, payment amount/status.
- Calculate number of unpaid orders.
- Summarize key metrics: total orders, revenue, AOV, success rate.

Sample Insights:

- Total Orders: 15,000
- Successful Payments: 13,800
- Orders Without Payment: 180

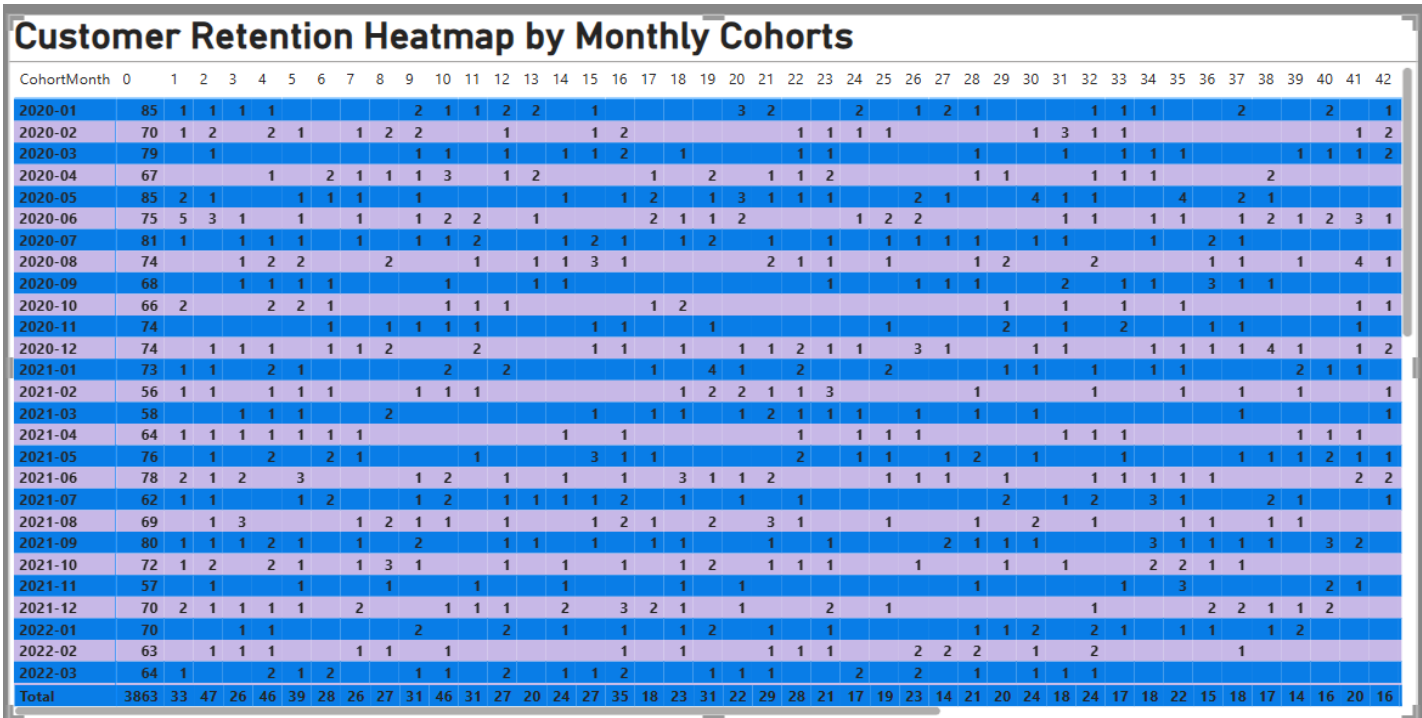
TASK 5: Customer Retention Analysis Report

Objective: Analyze customer retention patterns using cohort-based visualizations.

Data Source: customer_orders.csv

Attributes: customer_id, order_date, order_status, order_amount

Only records with order_status = 'Delivered' considered.



Methodology:

1. Cohort Definition: Based on first delivered purchase month.
2. Month Offset: Months after first purchase.
3. Retention Matrix: Created in Power BI with CohortMonth x MonthOffset.

Key Insights:

- Most customers drop off within 2-3 months.
- Cohorts like 2020-01 showed higher long-term retention.
- Newer cohorts show strong early engagement.